

Grandma's trauma – a critical appraisal of the evidence for transgenerational epigenetic inheritance in humans

By [Kevin Mitchell](#) - [May 29, 2018](#)

Can molecular memories of our ancestors' experiences affect our own behaviour and physiology? That idea has certainly grabbed hold of the public imagination, under the banner of the seemingly ubiquitous buzzword "epigenetics". [Transgenerational epigenetic inheritance](#) is the idea that a person's experiences can somehow mark their genomes in ways that are passed on to their children and grandchildren. Those marks on the genome are then thought to influence gene expression and affect the behaviour and physiology of people who inherit them.



The way this notion is referred to – both in popular pieces and in the scientific literature – you'd be forgiven for thinking it is an established fact in humans, based on mountains of consistent, compelling evidence. In fact, the opposite is true – it is based on the flimsiest of evidence from a very small number of studies with very small sample sizes and serious methodological flaws. [Note that there is, by contrast, very good evidence for this kind of mechanism in [nematodes](#) and [plants](#) and in specific circumstances involving transposable elements in [mice](#)].

To save you the trouble, I dig into the dismal details below. But first a quick tour of some recent articles in the popular press on the idea of ancestral epigenetic effects:

This one is from Discover magazine: [Grandma's Experiences Leave a Mark on Your Genes](#)

"Your ancestors' lousy childhoods or excellent adventures might change your personality, bequeathing anxiety or resilience by altering the epigenetic expressions of genes in the brain."

"According to the new insights of behavioral epigenetics, traumatic experiences in our past, or in our recent ancestors' past, leave molecular scars adhering to our DNA. Jews whose great-grandparents were chased from their Russian shtetls; Chinese whose grandparents lived through the ravages of

the Cultural Revolution; young immigrants from Africa whose parents survived massacres; adults of every ethnicity who grew up with alcoholic or abusive parents — all carry with them more than just memories.”

From TIME magazine: [Why Your DNA Isn't Your Destiny](#)

“The new field of epigenetics is showing how your environment and your choices can influence your genetic code — and that of your kids”

This recent one is from the New York Review of Books: [Epigenetics: The Evolution Revolution](#)

“This mechanism can be the hidden cause of our feelings of depression, anxiety, or paranoia. What is perhaps most surprising of all, this alteration could, in some cases, be passed on to future generations who have never directly experienced the stresses that caused their forebears’ depression or ill health.”

And this one is from ABC Science in Australia: [Epigenetics: how your life could change the cells of your grandkids](#)

“There's a very famous well-documented case where we can clearly see the impact of famine during pregnancy on a population over generations”, Professor Clark said. [Professor Susan Clark, Head of Genomics and Epigenetics at the Garvan Institute of Medical Research.] “In humans, the best example is during the WWII and the Dutch winter,” she said. During WWII, the Germans cut off food supplies to parts of the Netherlands causing a famine. Professor Clark said babies born to women during this time had a lower birthweight. When those babies grew up and had their own babies, the third generation had significantly more problems with diabetes and obesity than the rest of the population.”

There are dozens of others I could have chosen, from equally prominent titles. They almost all give the impression that the evidence for transgenerational epigenetic effects in humans is very strong, even if the underlying mechanisms remain mysterious. (Here is [an exception](#), by Adam Rutherford).

Many of them go further and claim that such findings have revolutionary implications, overturning Darwinian theories of evolution, refuting genetic determinism (a straw man), and implicating epigenetics as a crucial new mechanism in medicine and public health – both a cause of disease and a potential therapeutic target.

So, let’s take a look at some of these studies and see if the hype is warranted. (Spoiler: it isn’t).

Here is an early one, from 2006:

Sex-specific, male-line transgenerational responses in humans.

Pembrey ME1, Bygren LO, Kaati G, Edvinsson S, Northstone K, Sjöström M, Golding J; ALSPAC Study Team. [Eur J Hum Genet.2006 Feb;14\(2\):159-66.](#)

“We analysed food supply effects on offspring and grandchild mortality risk ratios (RR) using 303 probands and their 1818 parents and grandparents from the 1890, 1905 and 1920 Överkalix cohorts, northern Sweden... Sex-specific effects were shown in the Överkalix data; paternal grandfather’s food supply was only linked to the mortality RR of grandsons, while paternal grandmother’s food supply was only associated with the granddaughters’ mortality RR. These transgenerational effects were observed with exposure during the SGP [slow growth phase] (both grandparents) or fetal/infant life (grandmothers) but not during either grandparent’s puberty. We conclude that sex-specific, male-

line transgenerational responses exist in humans and hypothesise that these transmissions are mediated by the sex chromosomes, X and Y. Such responses add an entirely new dimension to the study of gene–environment interactions in development and health.”

A couple of things jump out here – first, the sample is **tiny**, for an epidemiological study – just 303 people. Second, the sex-specific effects were not specifically hypothesised – they just emerged from the data. They are also bizarrely arbitrary.

The authors found no general effect of grandparent’s nutrition during their slow growth phase (preteen years) on the mortality of their grandchildren. What do you do when you get no main effect? Arbitrarily test some covariates, of course, and in these studies, sex is the covariate that keeps on giving, especially because as you test it in combinations across generations it exponentially increases the hypothesis space that you can gratuitously explore. In this case, the probands’ paternal grandfather’s nutrition had an effect (and not any of their other grandparents) but only if the proband was male. And the paternal grandmother’s food supply had an effect but only if the proband was female.

Why? How? These are presented as interesting sex-specific effects and the authors hypothesise post hoc that they may involve epigenetic modifications of genes on the X and Y chromosomes, but really this is wild speculation. A more skeptical interpretation (appropriately so in my view) is that these “findings” are simply noise. They pop up as statistically significant amid a sea of non-significance, but they are in fact most likely just spurious statistical blips.

We will see this trend repeated over and over in other studies. Here’s another one:

Transgenerational effects of prenatal exposure to the Dutch famine on neonatal adiposity and health in later life. Painter RC, Osmond C, Gluckman P, Hanson M, Phillips DI, Roseboom TJ. [BJOG. 2008 Sep;115\(10\):1243-9.](#) doi: 10.1111/j.1471-0528.2008.01822.x.

OBJECTIVE: Maternal undernutrition during gestation is associated with increased metabolic and cardiovascular disease in the offspring. We investigated whether these effects may persist in subsequent generations. DESIGN: Historical cohort study. SETTING: Interview during a clinic or home visit or by telephone. POPULATION: Men and women born in the Wilhelmina Gasthuis in Amsterdam between November 1943 and February 1947. METHODS: We interviewed cohort members (F1) born around the time of the 1944-45 Dutch famine, who were exposed or unexposed to famine in utero, about their offspring (F2). MAIN OUTCOME MEASURES: Birthweight, birth length, ponderal index and health in later life (as reported by F1) of the offspring (F2) of 855 participating cohort members, according to F1 famine exposure in utero. RESULTS: F1 famine exposure in utero did not affect F2 (n = 1496) birthweight, but, among the offspring of famine-exposed F1 women, F2 birth length was decreased (-0.6 cm, P adjusted for F2 gender and birth order = 0.01) and F2 ponderal index was increased (+1.2 kg/m³, P adjusted for F2 gender and birth order = 0.001). The association remained unaltered after adjusting for possible confounders. The offspring of F1 women who were exposed to famine in utero also had poor health 1.8 (95% CI 1.1-2.7) times more frequently in later life (due to miscellaneous causes) than that of F1 unexposed women. CONCLUSIONS: We did not find transgenerational effects of prenatal exposure to famine on birthweight nor on cardiovascular and metabolic disease rates. F1 famine exposure in utero was, however, associated with increased F2 neonatal adiposity and poor health in later life. Our findings may imply that the increase in chronic disease after famine exposure in utero is not limited to the F1 generation but persists in the F2 generation.

Here's the table showing the data on which those findings are based:

Table 2. Birth and health characteristics of the F2 offspring of F1 participating women and men according to timing of prenatal exposure to the Dutch famine			
	Unexposed born before	Prenatal exposure to famine	Unexposed conceived after
Characteristics of F2 offspring of F1 women			
Number	243	401	212
Age (years)	34	33	32
Birthweight F2 (g)	3476	3484	3468
Birth length F2 (cm)	50.5	49.6*	50.1
Ponderal index F2 (kg/m ³)	26.6	27.8*	26.5
Prematurity F2 (%)	7	11	7
Died after birth F2 (%)	3	4	2
F2 poor health (%)	16	18	14
Congenital (%)**	1	1	2
Cardiovascular/metabolic (%)**	3	2	2
Psychiatric (%)**	4	2	2
Other (%)**	7	12*	9
Characteristics of F2 offspring of F1 men			
Number	197	256	187
Age (years)	31	30	29
Birthweight F2 (g)	3304	3298	3313
Birth length F2 (cm)	50.3	50.5	50.3
Ponderal index F2 (kg/m ³)	25.9	25.9	26.3
Prematurity F2 (%)	7	4	4
Died after birth F2 (%)	3	1	1
F2 poor health (%)	12	12	8
Congenital (%)**	4	4	1
Cardiovascular/metabolic (%)**	1	0	1
Psychiatric (%)**	1	1	1
Other (%)**	7	7	6

Again, a small sample, with lots of parameters studied (e.g., causes of death, where “Other” was the only category to show a significant effect, with a tiny number of people), with no particular hypotheses about which ones are expected to show an effect, in which direction. Basically, any difference anywhere will do.

The tiny differences in that table are taken as justifying the sweeping general claim made in the title of the paper. (And of course, many people will cite it based on the title, presuming the evidence actually supports such a claim).

An interesting point about this study is that the children of F1 males who were exposed to the famine conditions around the time of their birth showed no effect on any measure. But... wait for it... a follow-up study of the *exact same people* later in life found an “effect” on the children of F1 males but not those of F1 females:

Transgenerational effects of prenatal exposure to the 1944-45 Dutch famine.

Veenendaal MV¹, Painter RC, de Rooij SR, Bossuyt PM, van der Post JA, Gluckman PD, Hanson MA, Roseboom TJ. [BJOG. 2013Apr;120\(5\):548-53](#). doi: 10.1111/1471-0528.12136. Epub 2013 Jan 24.

OBJECTIVE: We previously showed that maternal under-nutrition during gestation is associated with increased metabolic and cardiovascular disease in the offspring. Also, we found increased neonatal adiposity among the grandchildren of women who had been undernourished during pregnancy. In the present study we investigated whether these transgenerational effects have led to altered body composition and poorer health in adulthood in the grandchildren. **DESIGN:** Historical cohort study. **SETTING:** Web-based questionnaire. **POPULATION:** The adult offspring (F2) of a cohort of men and women (F1) born around the time of the 1944-45 Dutch famine. **METHODS:** We approached the F2 adults through their parents. Participating F2 adults (n = 360, mean age 37 years) completed an online questionnaire. **MAIN OUTCOME MEASURES:** Weight, body mass index (BMI), and health in F2 adults, according to F1 prenatal famine exposure. **RESULTS:** Adult offspring (F2) of prenatally exposed F1 fathers had higher weights and BMIs than offspring of prenatally unexposed F1 fathers (+4.9 kg, P = 0.03; +1.6 kg/m², P = 0.006). No such effect was found for the F2 offspring of prenatally exposed F1 mothers. We observed no differences in adult health between the F2 generation groups. **CONCLUSIONS:** Offspring of prenatally undernourished fathers, but not mothers, were heavier and more obese than offspring of fathers and mothers who had not been undernourished prenatally. We found no evidence of transgenerational effects of grandmaternal under-nutrition during gestation on the health of this relatively young group, but the increased adiposity in the offspring of prenatally undernourished fathers may lead to increased chronic disease rates in the future.

Surely these kinds of studies are not all that bad, you say. Perhaps I'm picking some particularly egregious ones? Well, no. These are the ones that get cited all the time as the evidence for transgenerational effects of famine. And I've yet to find one that is in any way convincing.

Let's do another:

Change in paternal grandmothers' early food supply influenced cardiovascular mortality of the female grandchildren. Bygren LO, Tinghög P, Carstensen J, Edvinsson S, Kaati G, Pembrey ME, Sjöström M. [BMC Genet. 2014 Feb 20;15:12](#). doi: 10.1186/1471-2156-15-12.

Background: This study investigated whether large fluctuations in food availability during grandparents' early development influenced grandchildren's cardiovascular mortality. We reported earlier that changes in availability of food - from good to poor or from poor to good - during intrauterine development was followed by a double risk of sudden death as an adult, and that mortality rate can be associated with ancestors' childhood availability of food. We have now studied transgenerational responses (TGR) to sharp differences of harvest between two consecutive years for ancestors of 317 people in Överkalix, Sweden. **Results:** The confidence intervals were very wide but we found a striking TGR. There was no response in cardiovascular mortality in the grandchild from sharp changes of early exposure, experienced by three of the four grandparents (maternal grandparents and paternal grandfathers). **If, however, the paternal grandmother up to puberty lived through a sharp change in food supply from one year to next, her sons' daughters had an excess risk for cardiovascular mortality (HR 2.69, 95% confidence interval 1.05-6.92).** Selection or learning and imitation are unlikely explanations. X-linked epigenetic inheritance via spermatozoa seemed to be plausible, with the transmission, limited to being through the father,

*possibly explained by the sex differences in meiosis. **Conclusion:** The shock of change in food availability seems to give specific transgenerational responses.*

This is another orgy of covariate mining in a tiny sample. The great thing is that you can mine by sex combinatorially across generations, so you really get extra juice out of it for dredging for “significant” results somewhere. In this case, it is supposedly an effect only on the paternal grandmother that matters – so transmitted first through the female germline and then through the male germline, AND it only affects the granddaughters, not the grandsons! That’s some serious multiple testing, with no prior hypothesis, in a sample of 317 people!

There are a number of other studies along the same lines, including some looking at the supposed effects of things like grandparents smoking from an early age. They all suffer from the same problems:

1. Very small samples
2. Lack of predefined hypotheses
3. Extreme, combinatorial covariate dredging (i.e., massive multiple testing)
4. HARKing (hypothesising after results are known)

These all fall under the banner of [Questionable Research Practices](#) – the kinds of things that have filled the scientific literature in many fields with spurious findings and [false positives](#). This is the difference between wanting to test something (good science) and wanting to find something (bad science).

Taking whatever “significant” results pop up from these kinds of analyses at face value (as opposed to seeing them for noise) leads the authors to contort themselves into some truly arcane positions, like [this one](#): “*The evidence from this study suggests that when the mother does not smoke in pregnancy the maternal grandmother's smoking habit in pregnancy has a positive association with her grandson's fetal growth.*” Got that? Grandma’s smoking can have an effect on her daughter’s (not her son’s) sons (not daughters) but only if mom didn’t smoke herself.

These kinds of uber-specific scenarios are absurd on their face, to say nothing of the fact that they would require the invention of multiple new biological mechanisms to explain the sex-specific transmission (often switching from one sex to the other as it goes), as well as the sex-specific effects on grandchildren. They certainly don’t justify the sweeping generalisations made in the field, when the only way to get a significant result is to carve the data eight ways.

Things don’t get any better in recent papers that have attempted to identify the supposed genomic marks (thought to be mediated by DNA methylation) responsible for these supposed effects, like this one:

Grandmaternal stress during pregnancy and DNA methylation of the third generation: an epigenome-wide association study.

Serpeloni F, Radtke K, de Assis SG, Henning F, Nätt D, Elbert T. [Transl Psychiatry. 2017 Aug 15;7\(8\):e1202](#). doi: 10.1038/tp.2017.153.

Abstract: *Stress during pregnancy may impact subsequent generations, which is demonstrated by an increased susceptibility to childhood and adulthood health problems in the children and grandchildren. Although the importance of the prenatal environment is well reported with regards to future physical and emotional outcomes, little is known about the molecular mechanisms that mediate*

*the long-term consequences of early stress across generations. Recent studies have identified DNA methylation as a possible mediator of the impact of prenatal stress in the offspring. Whether psychosocial stress during pregnancy also affects DNA methylation of the grandchildren is still not known. In the present study we examined the multigenerational hypothesis, that is, grandmaternal exposure to psychosocial stress during pregnancy affecting DNA methylation of the grandchildren. **We determined the genome-wide DNA methylation profile in 121 children (65 females and 56 males) and tested for associations with exposure to grandmaternal interpersonal violence during pregnancy.** We observed methylation variations of five CpG sites significantly associated with the grandmother's report of exposure to violence while pregnant with the mothers of the children. The results revealed differential methylation of genes previously shown to be involved in circulatory system processes. This study provides support for DNA methylation as a biological mechanism involved in the transmission of stress across generations and motivates further investigations to examine prenatal-dependent DNA methylation as a potential biomarker for health problems.*

Yes, that's right – an [epigenome-wide association study](#) with a sample size of 121 – and the “cases” numbered 27. I'll just leave that there.

So, what are we to make of all this? You could be charitable and say the evidence is weak, circumstantial, observational, and correlative, and that it warrants circumspection and careful interpretation (and further research, of course!). I would go further and say that nothing in any of those papers rises to the level of what should properly be called a finding. There's no there there.

But wait, you say, what about all the animal studies that supposedly clearly show transgenerational epigenetic inheritance? Well, they suffer from all the same methodological problems as these human studies, as I have previously discussed [here](#) and [here](#).

How data become lore

So, if these data are so terrible, why do these studies get published and cited in the scientific literature and hyped so much in the popular press? There are a few factors at work, which also apply in many other fields:

1. The sociology of peer review. By definition, peer review is done by experts in “the field”. If you are an editor handling a paper on transgenerational epigenetic inheritance in humans (or animals), you're likely to turn to someone else who has published on the topic to review it. But in this case all the experts in the field are committed to the idea that transgenerational epigenetic inheritance in mammals is a real thing, and are therefore unlikely to question the underlying premise in the process of their review. [To be fair, a similar situation pertains in most fields].
2. Citation practices. Most people citing these studies have probably not read the primary papers or looked in detail at the data. They either just cite the headline claim or they recite someone else's citation, and then others recite that citation, and so on. It shouldn't be that way, but it is – people are lazy and trust that someone else has done the work to check whether the paper really shows what it claims to show. And that is how weak claims based on spurious findings somehow become established “facts”. Data become lore.

3. The media love a sexy story. There's no doubt that epigenetics is exciting. It challenges "dogma", it's got mavericks who buck the scientific establishment, it changes EVERYTHING about what we thought we knew about X, Y and Z, it's even got your grandmother for goodness sake. This all makes great copy, even if it's based on shaky science.

4. Public appetite. The idea of epigenetic effects resonates strongly among many members of the general public. This is not just because it makes cute stories or is scientifically unexpected. I think it's because it offers an escape from the spectre of genetic determinism – a spectre that has grown in power as we find more and more "genes for" more and more traits and disorders. Epigenetics seems to reassure (as the headline in TIME magazine put it) that DNA is not your destiny. That you – through the choices you make – can influence your own traits, and even influence those of your children and grandchildren. This is why people like [Deepak Chopra](#) have latched onto it, as part of an overall, spiritual idea of self-realisation.

So, there you have it. In my opinion, there is no convincing evidence showing transgenerational epigenetic inheritance in humans. But – for all the sociological reasons listed above – I don't expect we'll stop hearing about it any time soon.